



ThreatLens

Real-Time Code Security & Threat Intelligence Platform

3-Layer

Architecture

100ms

Detection Speed

10+

Vuln Patterns

Go + Python

Backend Stack

Team: Event Horizon

Hackathon 2025 · 100% Completion Milestone

Developers ship vulnerabilities daily — without knowing it.



Hardcoded Secrets

AWS keys, passwords, and tokens committed to repos — exposed in seconds by bots scanning GitHub.



Injections & XSS

SQL injections, shell commands, and unsanitized user input silently waiting to be exploited.



Malicious IOCs

Suspicious IPs, C2 domains, and hashes embedded in code with no visibility into threat reputation.

Three-Layer Intelligence Stack



Frontend

Electron + React

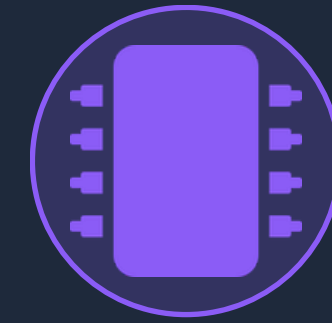
- ▶ Monaco Editor
- ▶ Debounced Analysis
- ▶ Native Diagnostics
- ▶ Dynamic Verdict Cards



Backend

Python FastAPI

- ▶ Regex Vuln Engine
- ▶ IOC Extractor
- ▶ SQLite Auth (RBAC)
- ▶ async httpx Proxy



Microservice

Go :8081

- ▶ Goroutine Concurrency
- ▶ 24-hr SQLite Cache
- ▶ Mock Threat Engine
- ▶ VirusTotal + AbuseIPDB

frontend

File Edit View Window

ThreatLens

```
2 import requests
3 import json
4
5 def initialize_cloud_client():
6
7     aws_access_key = "AKIA1234567890ABCDEF"
8
9     client = boto3.client(
10         's3',
11         aws_access_key_id=aws_access_key,
12
13         password = "super_secret_admin_pass"
14     )
15     return client
16
17 def fetch_external_config():
18
19     malicious_endpoint = "http://45.33.32.156/payload/config.json"
20
21     try:
22         response = requests.get(malicious_endpoint)
23         return response.json()
24     except Exception as e:
25         print(f"Failed to fetch config: {e}")
26         return None
27
28 if __name__ == "__main__":
29     cloud_client = initialize_cloud_client()
30     config = fetch_external_config()
31     print("Application initialized.")
32
```

LIVE DETECTION

Instant Vulnerability Identification

- CRITICAL

Line 7

AWS Key Exposed
- CRITICAL

Line 13

Hardcoded Password
- CRITICAL

Line 19

Malicious IP (88/100)
- MEDIUM

Line 19

Suspicious URL/Domain

Go Microservice · Concurrent IOC Enrichment

CRITICAL

Line 19

Found suspicious IP address

45.33.32.156

Verdict: **MALICIOUS**

Threat Score (88/100)

Reported 847 times for SSH brute force — flagged by 12 vendors.

Source: VirusTotal + AbuseIPDB

MEDIUM

Line 19

Found embedded URL/Domain

http://45.33.32.156/payload/config.json

Verdict: **BENIGN**

Threat Score (2/100)



Goroutine Concurrency

Spawns parallel goroutines per IOC via `sync.WaitGroup` — 50 simultaneous lookups without blocking.



24-Hour Caching Layer

No-CGO SQLite cache (modernc.org/sqlite). Instant replay for known artifacts under 24hrs old.



Deterministic Mock Engine

Injects 1-3s latency simulation. Returns hardcoded scores for demo IPs like 45.33.32.156 (88/100).



Dynamic Severity Upgrade

Medium → Critical auto-promotion when Go flags IOC as malicious. Monaco markers update instantly.

Premium Design System · Electron + React



Monaco Editor

VS Code-grade editing with native squiggle markers via setModelMarkers API.



Reactive Polling

useEffect watches IOCs and auto-fires /enrich requests on new IP/URL/Hash detection.



Threat Score Cards

0-100 animated progress bars, color-coded verdicts, spinner states during live lookups.



Scan History Strip

Horizontal scrollable Set of last 5 artifacts. One-click Clear resets all state and markers.

ThreatLens

```
1 // 3. Cross-Site Scripting (XSS)
2
3 app.post('/comment', (req, res) => {
4   const comment = req.body.comment;
5
6   res.send(
7     <h1>User Comment</h1>
8     <div>${comment}</div>
9   );
10 });
```

Python Regex Pattern Library · FastAPI /analyze

CRITICAL

AWS Access Keys

AKIA[0-9A-Z]{16}

CRITICAL

Hardcoded Passwords

(?i)password\s*=\s*['"]([^\"]+)[\"']

CRITICAL

SQL Injection

SELECT.*FROM|UNION ALL SELECT

CRITICAL

eval() / Shell Injection

\beval\s*\(|os\.system|subprocess\.Popen

LOW

Risky Bind Address

\b0\.0\.0\.0\b

IOC

IPv4 Extractor

(?:[0-9]{1,3}\.){3}[0-9]{1,3} (no local)

IOC

URL / Domain Extractor

http[s]?://[^\s]+

IOC

Hash Extractor (MD5/SHA256)

[a-f0-9]{32} | [a-f0-9]{64}

Three Purpose-Built Pitch Scripts



demo1_secrets.py

Secrets & Injections

- ✓ AWS AKIA key → CRITICAL
- ✓ Hardcoded password → CRITICAL
- ✓ SQL SELECT/UNION → CRITICAL

Python Regex Engine

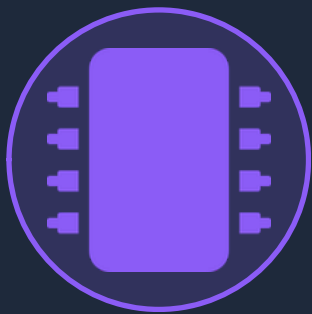


demo2_malicious_domain.py

Malicious Domain Intel

- ✓ Embedded http:// URL → IOC
- ✓ Go microservice enrichment
- ✓ Vendor verdict + threat score

Go Threat Intel Service



demo3_c2_ip_hash.py

C2 IP + Hash Detection

- ✓ 45.33.32.156 → 88/100 SSH BF
- ✓ SHA256 hash extraction
- ✓ Medium → CRITICAL upgrade

Dynamic Severity System